

Quick Reference Guide

Module 1: Introduction to the Artificial Intelligence Design Process

Learning Outcomes:

- Evaluate NLP applications for optimal results.
- Analyze an AI-driven product or process to determine how effectively it leverages data strategies to create and sustain a competitive advantage.
- Evaluate the cost metrics and contingency planning of an AI software development plan.
- Describe how AI cancers can affect AI applications.
- Identify the stages of the AI design process model.
- Appraise the business and technical requirements for a desired AI behavior.

Introduction to Artificial Intelligence

AI is the field of computer science focused on creating systems that mimic human intelligence—such as learning, reasoning, and language use. The AI ecosystem includes several key areas:

- Machine learning (ML): Enables machines to learn from data with minimal programming. Includes supervised, unsupervised, and semi-supervised approaches.
- Deep learning (DL): Uses multilayer neural networks to detect complex patterns; powers advances in computer vision and natural language processing.
- Computer vision (CV): Allows machines to interpret images and video for tasks like recognition, detection, and classification.
- Natural language processing (NLP): Enables computers to analyze, understand, and generate human language.
- Large language models (LLMs): Transformer-based systems trained on massive text data; used for generation, translation, and question-answering.
- Retrieval-augmented generation (RAG): Combines document retrieval with LLMs to deliver accurate, up-to-date, and cited responses.

- Multimodal AI: Integrates multiple data types (e.g., text, images, audio) to enrich human-like interaction.
- Automation agents: AI systems that plan and act toward goals using memory, tools, and feedback loops.

Stages of the AI Design Process

- Intelligence: The first stage is to identify the desired behavior of the AI product. This involves addressing two challenging choices – performance metric and scope.
 - Performance metric choice involves understanding how the performance frontier may be evolving in your chosen domain, so you can send targets for the AI design.
 - The other choice is to understand the scope of what you are asking AI to achieve. What is the intelligence you need in your application?
- Business process: The second stage is to find the business process that AI can support. To find such a business process, there are two key issues that need to be addressed. The first is the strategic consideration that involves deciding how to use AI to compete in the marketplace and the role that you want AI to play in your design process. The Delta model developed by Professor Hax at MIT Sloan suggests that there are three strategies to compete.
 - The first strategy is to be the best product player. In the case of AI, this means having the best technology possible.
 - The second strategy accordingly to the Delta model is to be a full customer solutions player. In the case of AI, this means providing a product that incorporates all AI and all the associated features that will make it a useful product or service.
 - The third strategy of the model is network externalities. The focus is on building the largest user base possible in a way that results in user getting more benefits as the size of the database increases.

The second consideration to be addressed while exploring a business process that AI can support is operational consideration. This involves specifying the business process where AI

will intervene in setting performance targets for such process, including not just AI but other collateral assets. The idea here is to focus on the process that will be impacted by AI so that it can give the adequate information on what is expected of the AI implementation. You need to define a business process and how AI will improve its performance targets.

- AI technology: The third stage focuses on AI technology choices and there are two key decisions to be made.
 - The first involves selecting the right intellectual property approach.
 - The second centres on picking the right data strategy.
- Tinkering: The fourth stage of the AI design process centers on tinkering with technology. In this stage also, there are two choices to make. The first one is the software development approach or strategy, and it needs to allow room for pivots. As the frontier of AI is rapidly moving, flexibility must be built into the software development process and adaptability incorporated into the design process. The more complex mechanical processes get, the more sophisticated the associated software tinkering will need be. There are four key points in a software development plan.
 - Code and data repository methodology which includes establishing where the different versions of the code will be kept, and what tools to do so will be used.
 - Testing and experimental plan that addresses how are you going to test the software and how will you manage the different issues that arise from the testing plan?
 - Graphic processing unit (GPU) experimental set up that rationalizes the use of resources. The experimental setup should be such that the use of the available machines is maximized and optimized.
 - Plan a budget to justify the investment as new software development projects will cost two and a half times more than what you think they will cost, and budgets always runs over.

The second choice in this stage of the design process is related to AI cancers and limitations. AI has some known issues that you will have to plan for and deal with.

- The first one are adversarial attacks. Research has shown that you can fool any deep

learning AI algorithm with simple modifications in the input signal.

- The second cancer is lack of generalization. In many cases, AI cannot generalize properly.
- The third cancer is bias. Machine learning algorithms often have the bias of the data that one uses to train them.
- Another related cancer is explainability. One of the big issues with modern deep learning algorithms is that they do not necessarily provide any insight into how they reach a conclusion.
- The fifth cancer is unintended behavior.

